Project Proposal - Week 1

Group	Group 2 - AI Exam Answer Grader
Roll Numbers	J008, J009, J010, J011, J012
Class & Course	B.Tech DS 4th year – Speech & Natural Language Processing (SNLP)

1. Problem

Faculty spend a substantial amount of time manually evaluating descriptive/subjective examination answers against a reference answer sheet and marking rubric. This manual process is slow, which doesn't scale with class size, and can produce inconsistent scores across evaluators or across time. There is a need for a component that can assist (not replace) this evaluation process with a consistent, explainable, and fast first-pass assessment.

2. Users

Primary users would be the faculty members grading descriptive exam or assignment answers, who need a fast, consistent first-pass score and feedback suggestions they can review and override. Secondary users are mainly students, who could receive practice feedback on draft answers before submission. Institutionally, this component is designed to plug directly into the AI University Platform's Academic Services layer, consuming structured answer text from Group 1 (Handwritten Examination Digitiser) as an upstream data source.

3. Input

- Question text
- Reference/model answer
- Marking rubric
- Student's answer
- Maximum marks for the question

4. Output

The output would be a structured JSON object.
For example:
{ "question": "Q4", "score": 7, "maximum_score": 10, "feedback": "The answer correctly explains...", "confidence": 0.87 }

5. NLP Task Formulation

The core problem is formulated as a semantic textual similarity / short-answer scoring task: given a student answer and a reference answer (plus rubric), predict a score (regression) or score band (ordinal classification), optionally supported by concept/keyword coverage against the rubric to generate explainable feedback.

6. Proposed Approaches

- Approach A (Baseline): TF-IDF / embedding-based cosine similarity between student answer and reference answer calibrated to a marks scale.
- Approach B (Improved): LLM or sentence-transformer-based rubric evaluation, scoring against individual rubric concepts rather than only overall similarity.

Both approaches will be implemented and compared quantitatively.

7. Dataset

Primary source: Our own class's mid-term/assignment submissions using existing rubrics and marks already assigned by faculty as ground truth.

Fallback/supplementary source: The public **ASAP-SAS** (Automated Student Assessment Prize - Short Answer Scoring) dataset, a standard benchmark for short-answer scoring.

Other possible datasets that can be utilised for this project are the **Mohler ASAG Dataset**, **SciEntsBank** and **EngSAF**.

If neither is sufficient in time, a small synthetic labelled set will be created by having group members grade sample answers against a rubric. All data handling will follow the privacy guidelines. No personally identifiable student information will be uploaded to public AI services, and any real institutional data used will be anonymised.

8. Expected Institutional Value

Reduces faculty grading time and effort on descriptive answers, improves consistency of scoring across large classes, gives students faster feedback, and demonstrates cross-group reusability by integrating Group 1's structured OCR output as an input source, which directly supports the AI University Platform.